Electrical and Computer Engineering
University of Thessaly

ECE327 - Digital Systems VLSI

DC and Transient Analysis of NMOS/PMOS Transistors and CMOS Inverters using SPICE Level 3 Models

Winter Semester — 2022-2023

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Contents

Exercise 1

(a)

Utilizing the **NGSPICE** simulation environment, the following current-voltage characteristics (I_{ds}) were obtained for both transistor types (NMOS and PMOS)(Figures 1,2,3,4). These characteristics illustrate the drain-to-source current as a function of both the drain-to-source voltage (V_{ds}) and the gate-to-source voltage (V_{gs}).

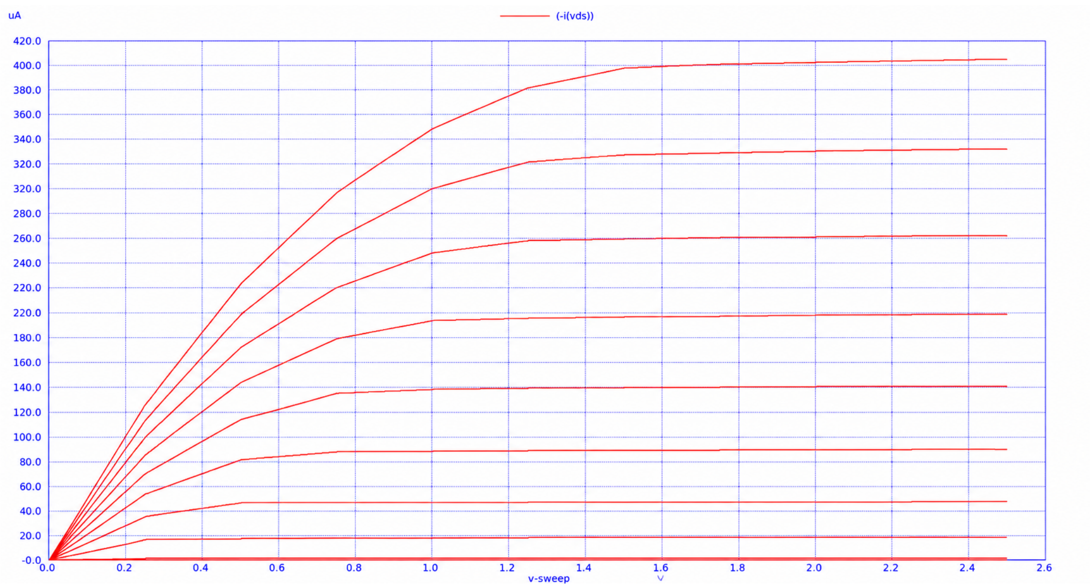


Figure 1: NMOS drain current (I_{ds}) characteristics as a function of drain-to-source voltage (V_{ds}) for various gate-to-source voltage (V_{gs}) values.

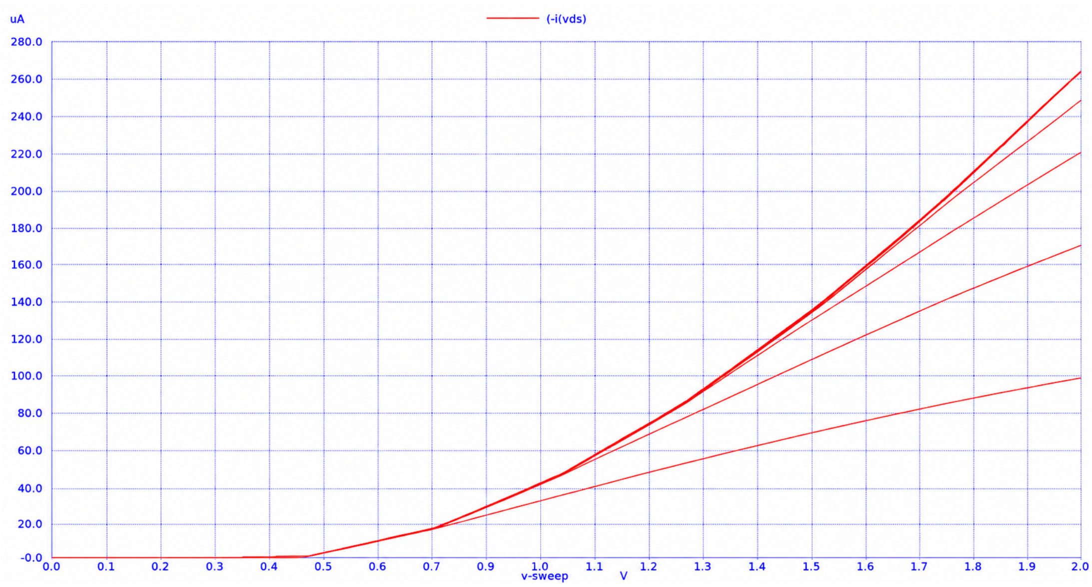


Figure 2: NMOS drain current (I_{ds}) characteristics as a function of gate-to-source voltage (V_{gs}) for various drain-to-source voltage (V_{ds}) values.

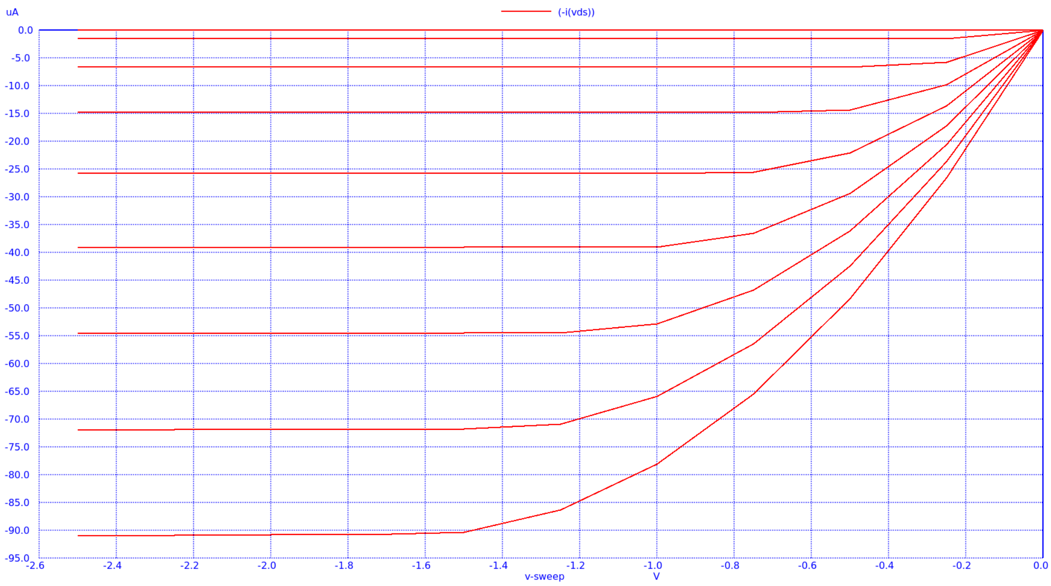


Figure 3: PMOS drain current (I_{ds}) characteristics as a function of drain-to-source voltage (V_{ds}) for various gate-to-source voltage (V_{gs}) values.

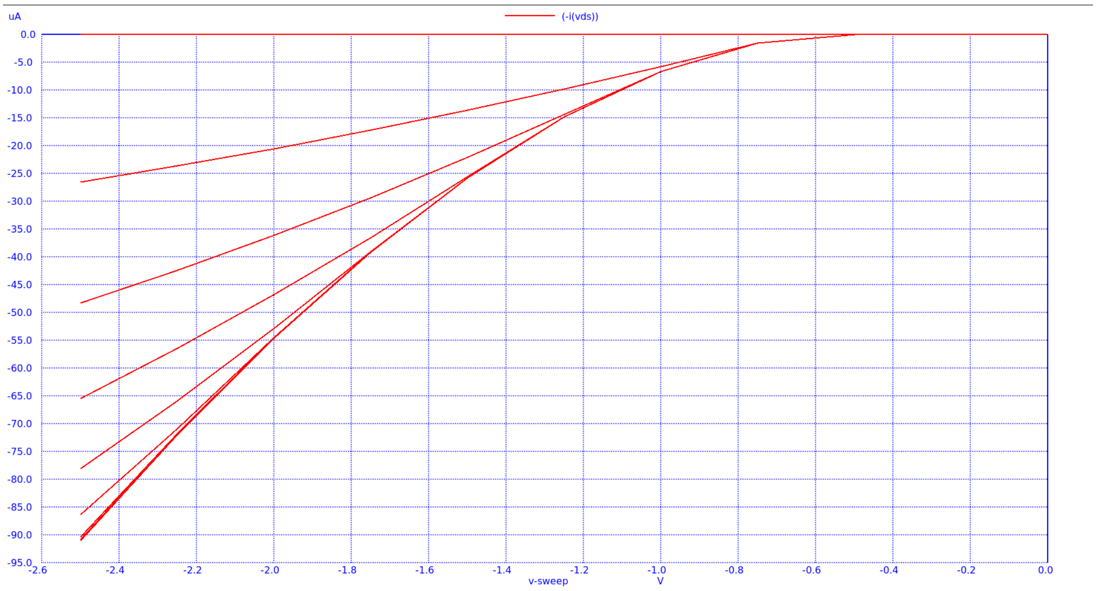


Figure 4: PMOS drain current (I_{ds}) characteristics as a function of gate-to-source voltage (V_{gs}) for various drain-to-source voltage (V_{ds}) values.

Figure 5: PMOS transistor drain current characteristics: (a) Output characteristics and (b) Transfer characteristics.

The following figures (6,**??**), generated via the LTspice graphical interface, illustrate the current-voltage ($I - V$) characteristics for both transistor types as a function of the drain-to-source voltage (V_{ds}).

A key aspect of this analysis is the identification of the saturation points, defined by the condition $V_{ds} = V_{gs} - V_t$. These points are marked on the curves for various V_{gs} levels to delineate the transition from the linear (triode) region to the saturation region. Furthermore, vertical projections are utilized to measure the vertical distances between successive I_{ds} curves, facilitating a comparative analysis of the current increment relative to the gate-to-source voltage steps.

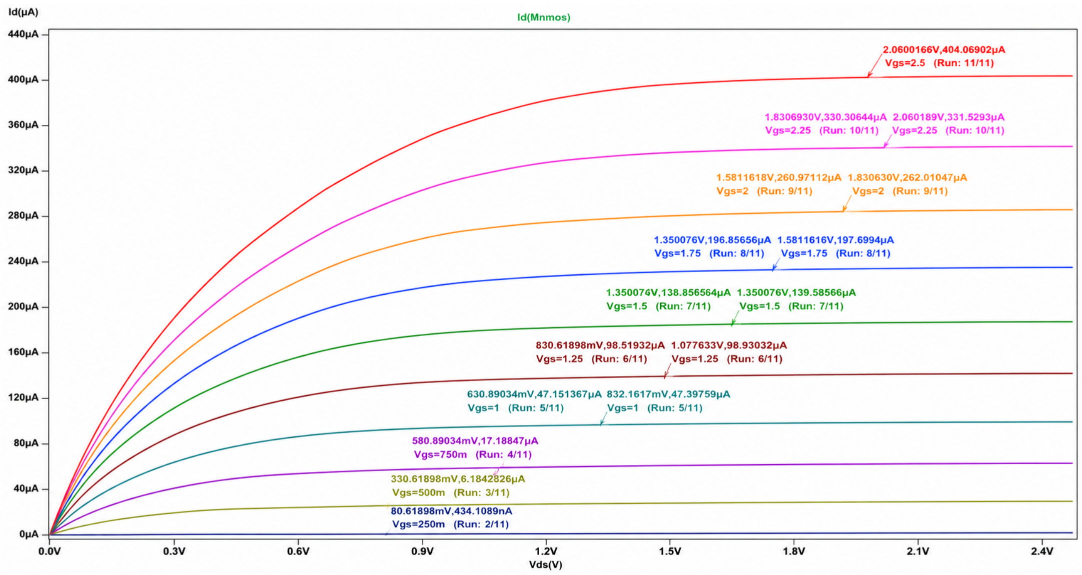


Figure 6: Saturation points of the nMOS transistor

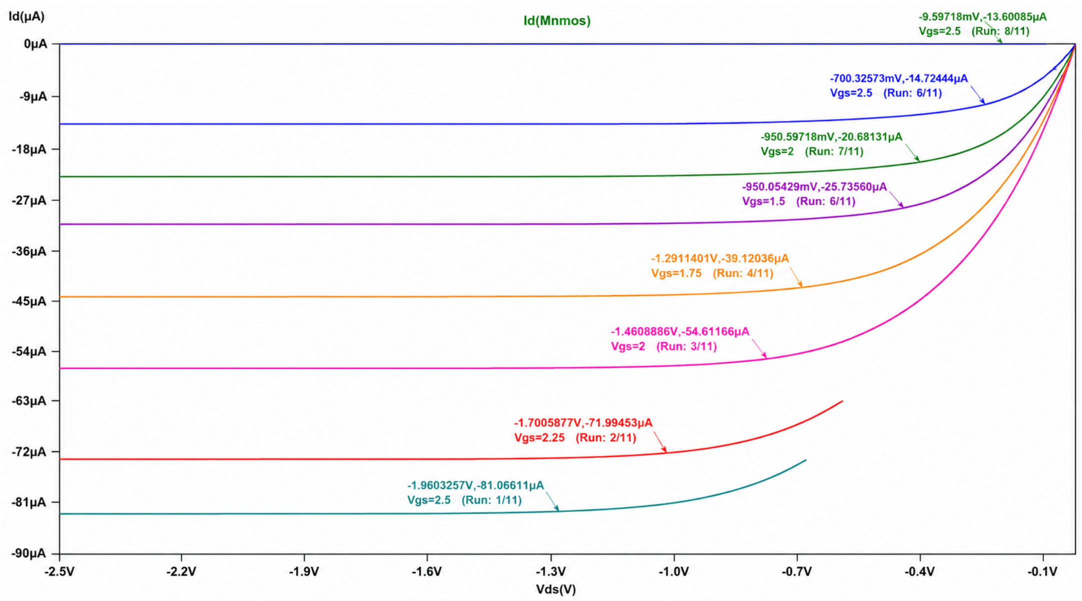


Figure 7: Saturation points of the pMOS transistor

The following tables present the extracted saturation points for both NMOS and PMOS transistors. These data points correspond to various gate-to-source voltage (V_{gs}) levels, providing a quantitative overview of the transition boundaries for each device type.

Table 1: NMOS Saturation Points for various V_{GS} levels ($V_t = 0.42V$).

V_{GS} **(V)**	$V_{DS,sat} = V_{GS} - V_t$ **(V)**
0V	OFF ($V_{GS} < V_t$)
0.25V	OFF ($V_{GS} < V_t$)
0.5V	0.08V
0.75V	0.33V
1V	0.58V
1.25V	0.83V
1.5V	1.08V
1.75V	1.33V
2V	1.58V
2.25V	1.83V
2.5V	2.08V

Table 2: PMOS Saturation Points for various V_{GS} levels ($V_t = -0.55V$).

V_{GS} **(V)**	$V_{DS,sat} = V_{GS} - V_t$ **(V)**
-2.5V	-1.95V
-2.25V	-1.7V
-2V	-1.45V
-1.75V	-1.2V
-1.5V	-0.95V
-1.25V	-0.7V
-1V	-0.45V
-0.75V	-0.2V
-0.5V	OFF ($V_{GS} > V_t$)
-0.25V	OFF ($V_{GS} > V_t$)
0V	OFF ($V_{GS} > V_t$)

For NMOS transistors, the set of points on each characteristic curve satisfying the condition $V_{ds} < V_{gs} - V_t$ defines the linear (triode) region. In this region, the drain current increases proportionally with both V_{ds} and V_{gs}. Under these conditions, the transistor behaves similarly to a voltage-controlled resistor, where the current varies linearly with the drain-to-source voltage.

Additionally, the curves where $I_d = 0$ correspond to the cut-off (OFF) region. When $V_{gs} \leq V_t$, the current between the drain and the source (I_d) is practically negligible, and the transistor is considered to be in the non-conducting state.

The region to the right of the saturation points is defined as the saturation region. For all points in this zone, the condition $V_{ds} \geq V_{gs} - V_t$ holds. In saturation, the transistor operates in a state where the drain current I_d is primarily controlled by the gate voltage and remains significantly independent of the drain-to-source voltage.

These observations are equally applicable to PMOS transistors, with the difference that the polarity of voltages and currents is reversed; consequently, the corresponding inequality relationships and operating intervals are inverted.

The expression of the ideal drain current in the saturation region, given by $I_D = \frac{k_n}{2}(V_{GS} - V_t)^2$, implies that the current remains constant and independent of the drain-to-source voltage. However, as observed in the simulation results, the current in the saturation region does not remain perfectly stable; instead, it exhibits a slight increase with rising V_{DS}. This indicates that I_D is not entirely independent of the drain voltage.

This physical phenomenon occurs because an increase in V_{DS} causes the depletion region at the drain-substrate junction to expand towards the channel, effectively reducing the length of the electrical channel (L). A more accurate representation of the drain current is provided by the following equation:

$$I_D = \frac{k_n}{2}(V_{GS} - V_t)^2(1 + \lambda V_{DS}) \tag{1}$$

where λ represents the hannel-length modulation coefficient, which is inversely proportional to the channel length.

Using the LTspice simulation environment LTspice, we measured the vertical distances between successive drain current characteristic curves (I_{ds}) for various values V_{gs}, plotted as a function of V_{ds}.

For the NMOS transistor, the following observations were made (table 4):

Table 3: Vertical distances between successive I_{ds} characteristic curves for the nMOS transistor.

V_{gs1} **(V)**	V_{gs2} **(V)**	**d (**μA**)**
0.5	0.75	15
0.75	1	30
1	1.25	41
1.25	1.5	50
1.5	1.75	57
1.75	2	63
2	2.25	68
2.25	2.5	72

Similarly, the vertical distances between the successive I_{ds} characteristics in the case of the pMOS transistor are summarized in table **??**

It is observed that the saturation current (I_D) for both NMOS and PMOS devices exhibits a quadratic dependence on the gate-to-source voltage (V_{GS}). Specifically, for every constant

Table 4: Vertical distances between successive I_{ds} characteristic curves for the nMOS transistor.

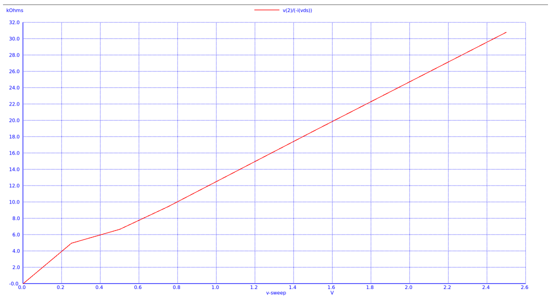
V_{gs1} **(V)**	V_{gs2} **(V)**	**d** (μA)
-2.5	-2.25	-19
-2.25	-2	-17
-2	-1.75	-15
-1.75	-1.5	-12
-1.5	-1.25	-10
-1.25	-1	-7
-1	-0.75	-3.4
2.25	2.5	72

increment of V_{GS}, the drain current increases near-quadratically, which explains why the vertical distance between successive V_{GS} curves is not constant but increases progressively.

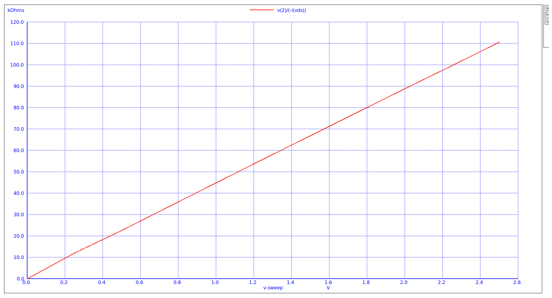
This $I_D \propto V_{GS}^2$ relationship is more pronounced in transistors with a large channel length (L). Conversely, in short-channel devices, the physical phenomenon of **velocity saturation** becomes dominant. In such cases, the dependence of the drain current on V_{GS} deviates from the square law and tends to become linear, significantly affecting the device's transconductance and overall switching behavior.

To evaluate the absolute (instantaneous) resistance (R_{eqabs}) of the transistors, the ngspice simulator was used to generate plots that illustrate the resistance values across an input voltage sweep from 0V to 2.5V. This analysis accounts for the corresponding drain current at specified voltage levels from source to gate ($V_{gs} \in \{0.8, 1.2, 2, 2.5\}V$).

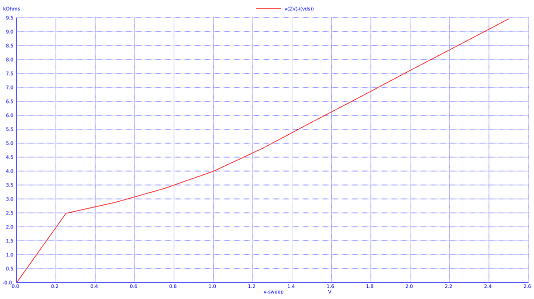
For the PMOS devices, given the inverted polarity of the voltages and currents, the characteristics are symmetrically reversed compared to the NMOS. The resulting graphical representations of the absolute resistance for both NMOS and PMOS transistors are presented below, providing a clear visualization of how the device resistance modulates as a function of the operating point (8,9).



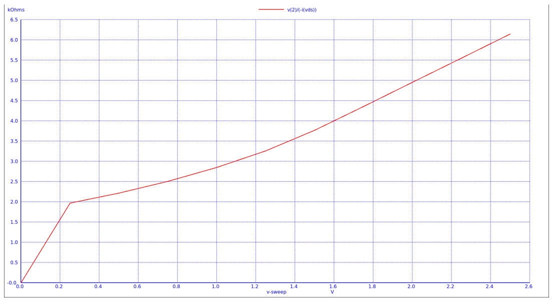
(a) Absolute resistance of nmos transistor and $V_{gs} = 0.8V$



(b) Absolute resistance of nmos transistor and $V_{gs} = 1.2V$

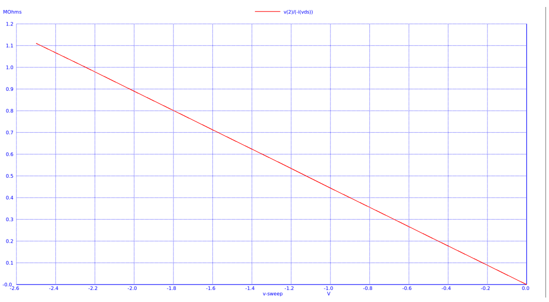


(c) Absolute resistance of nmos transistor and $V_{gs} = 2V$

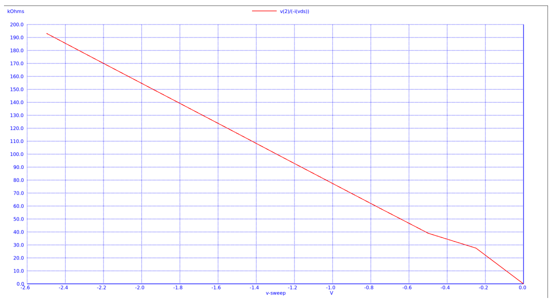


(d) Absolute resistance of nmos transistor and $V_{gs} = 2.5V$

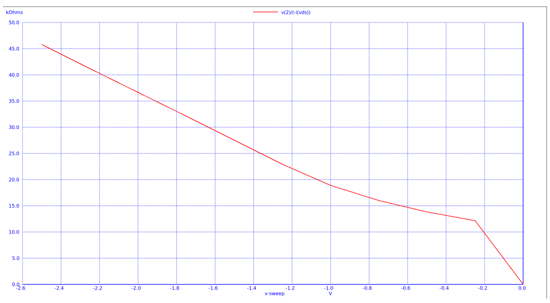
Figure 8: NMOS transistor resistance plots.



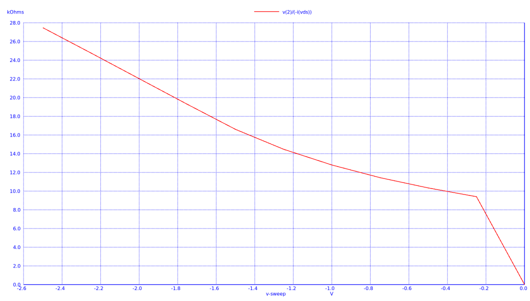
(a) Absolute resistance of pmos transistor and $V_{gs} = -0.8V$



(b) Absolute resistance of pmos transistor and $V_{gs} = -1.2V$



(c) Absolute resistance of pmos transistor and $V_{gs} = -2V$



(d) Absolute resistance of pmos transistor and $V_{gs} = -2.5V$

Figure 9: PMOS transistor resistance plots.

Based on the simulation results, it is observed that for NMOS devices, the instantaneous resistance increases proportionally with the drain-to-source voltage (V_{ds}). Similarly, for PMOS devices, the resistance increases as the absolute value of V_{ds} rises.

To determine the average equivalent resistance (R_{eqav}) for each transistor, we utilized the simulated characteristic plots across the full input voltage swing. Specifically, the analysis was conducted for a voltage range from 0V to 2.5V for the NMOS devices, and correspondingly from -2.5V to 0V for the PMOS devices, to account for their complementary nature.

The R_{eqav} is calculated by averaging the instantaneous resistance values over the operating interval, providing a more reliable metric for dynamic performance analysis compared to static measurements. The extracted mean values from the aforementioned simulation plots for both nmos and pmos transistors are summarized in the following tables (56, illustrating the impact of V_{gs} modulation on the effective channel resistance.

Table 5: Equivalent Resistance (R_{eq}) values for NMOS transistor.

V_{GS} **(V)**	R_{eq} **($k\Omega$)**
0.8V	55.7
1.2V	15.7
2V	5.05
2.5V	3.41

Table 6: Equivalent Resistance (R_{eq}) values for PMOS transistor.

V_{GS} **(V)**	R_{eq} **($k\Omega$)**
-0.8V	555.52
-1.2V	97.3
-2V	24.27
-2.5V	15.31

It is observed that as the absolute value of the gate-to-source voltage ($|V_{gs}|$) decreases, the average resistance (R_{eqav}) of the devices increases significantly.

Furthermore, a distinct performance disparity is noted between the two device types: the average resistances of the PMOS transistors are markedly higher than those of the NMOS transistors for equivalent operating conditions.

An additional analysis was performed to determine the average equivalent resistance (R_{eq}) specifically within the saturation regions of the NMOS and PMOS transistors. The saturation region for each gate voltage level was defined starting from the pinch-off point ($V_{DS} \geq V_{GS} - V_t$) up to the maximum supply voltage of 2.5V (or -2.5V for PMOS).

Applying the same methodology as before, we estimate the average resistance within the range from the saturation point until the supply voltage for every one of the 8 different scenarios.The results of our research are summarized in the table 7 for both transistors.

Table 7: Average Equivalent Resistance (R_{eq}) in the Saturation Region.

Type	V_{GS} **(V)**	**Saturation Interval (V)**	R_{eq} **($k\Omega$)**
NMOS	0.8V	0.38V to 2.5V	61.5
	1.2V	0.78V to 2.5V	19.0
	2V	1.58V to 2.5V	7.45
	2.5V	2.08V to 2.5V	5.53
PMOS	-0.8V	-2.5V to -0.25V	611.0
	-1.2V	-2.5V to -0.65V	126.0
	-2V	-2.5V to -1.45V	36.5
	-2.5V	-2.5V to -1.95V	24.55

Consistent with our previous findings, the data confirms that the resistance remains significantly higher at lower $|V_{gs}|$ levels and decreases as the transistor is driven more strongly. Furthermore, the PMOS device continues to exhibit higher resistance values compared to the NMOS across all saturation intervals, reinforcing the impact of lower hole mobility on device performance.

(b)

To calculate the average equivalent resistance (R_{eq}) for both NMOS and PMOS transistors, we utilized the mathematical relationship between channel resistance and propagation delay. Specifically, the equivalent resistance $R_{eq,p}$ is derived from the low-to-high transition delay (t_{plh}), while $R_{eq,n}$ is determined via the high-to-low transition delay (t_{phl}).

Using the ngspice simulator, we implemented a transient analysis involving a load capacitor ($C = 0.1\,pF$). The capacitor is first discharged through an NMOS transistor to simulate a logic 1 to logic 0 transition, and subsequently recharged through a PMOS transistor to simulate a logic 0 to logic 1 transition.

The charging and discharging procedure is shown in figures 10,11.

The charging delay (t_{plh}) is defined as the time interval required for the capacitor voltage to rise from 0V to the 50% threshold of 1.25V. Conversely, the discharging delay (t_{phl}) is defined as the time required for the capacitor voltage to drop from the supply level of 2.5V to 1.25V.By measuring the resulting propagation delays in the figures 10 and 11 we can apply the standard delay formulas to extract the effective switching resistance of each device.

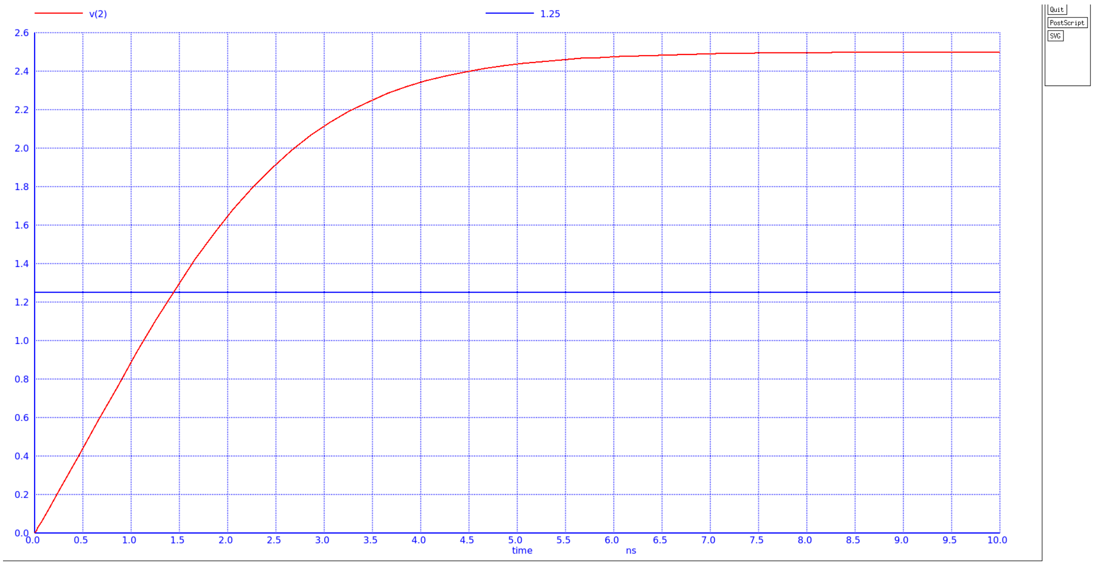


Figure 10: Charging procedure of the capacitor

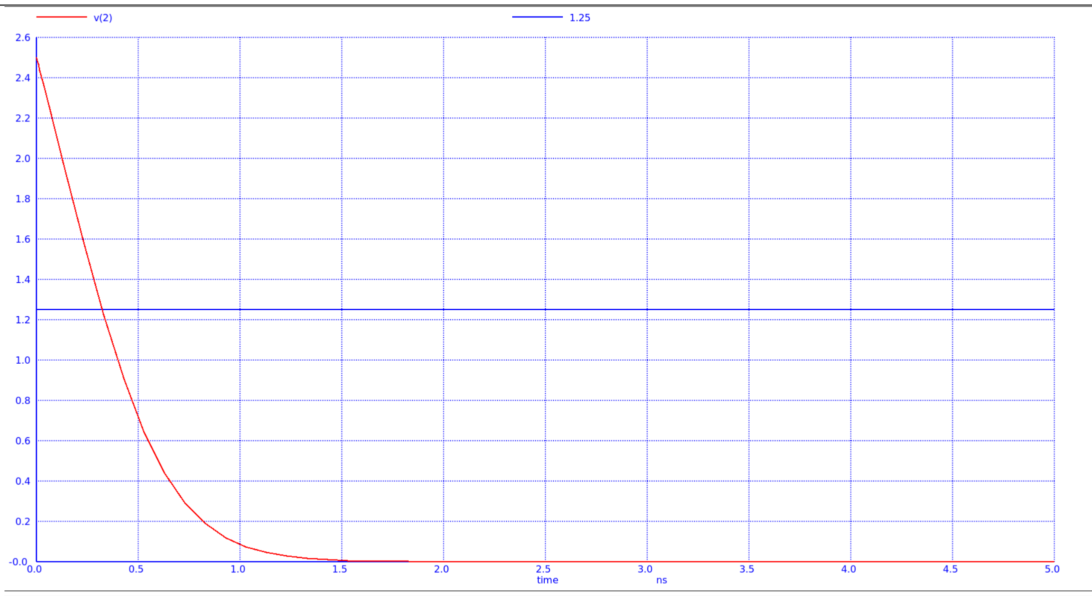


Figure 11: Discharging procedure of the capacitor

Based on the transient analysis, the measured propagation delays are:

- $t_{pHL} = 0.321\,ns$
- $t_{pLH} = 1.46\,ns$

Using the analytical delay model $t_p = \ln(2) \cdot R_{eq} \cdot C_L$, we extract the equivalent resistances for a load capacitance of $C_L = 0.1\,pF$:

$$R_{eqn} = 4.652\,k\Omega, \quad R_{eqp} = 21.15\,k\Omega \tag{2}$$

According to the results we could conclude that the PMOS resistance is significantly higher than that of an NMOS of identical dimensions, as evidenced by the notably longer charging time (t_{pLH}) compared to the discharging time (t_{pHL}). These values, derived from dynamic transient analysis, are consistent with the average resistance results obtained through the previous DC analysis.

Exercise 2

Utilizing the simulator, we configured a circuit consisting of two NMOS transistors connected in series with the supply voltage ($V_{DD} = 2.5V$). Both output nodes (Node 2 and Node 4) were initialized with zero voltage to observe the transient charging behavior. Upon application of the input signal ($V_{in} = 2.5V$) to the gates of both devices, the voltages at these nodes were measured to evaluate the cumulative voltage drop.

The circuit implementation is illustrated in Figure 12. As the simulation progresses, the output voltage fails to reach the full rail value of $2.5V$. This phenomenon is attributed to the fact that an NMOS transistor can only pass a maximum voltage of $V_G - V_T$. Furthermore, the second transistor (M_2) exhibits a more pronounced voltage drop due to the Body Effect, as its source potential rises relative to the grounded substrate, thereby increasing its effective threshold voltage ($V_{T,eff}$).

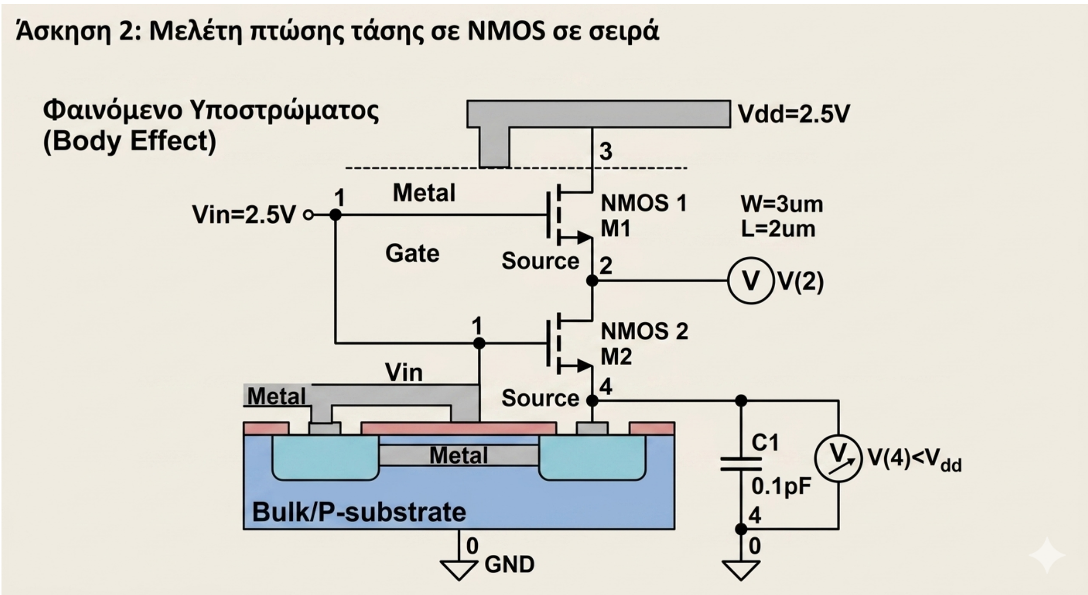


Figure 12: nMOS transistors connected in series with a capacitor

Based on the transient analysis in ngspice, we observe the following:

- Node 2 Voltage:Stabilizes at 1.7V(13) The total voltage drop is $0.8V$. Although the nominal V_{T0} is $0.42V$, the Body Effect increases the effective threshold voltage (V_T) because the source potential rises while the bulk remains grounded.
- Node 4 Voltage: Stabilizes at approximately 1.7V (equal to Node 2).

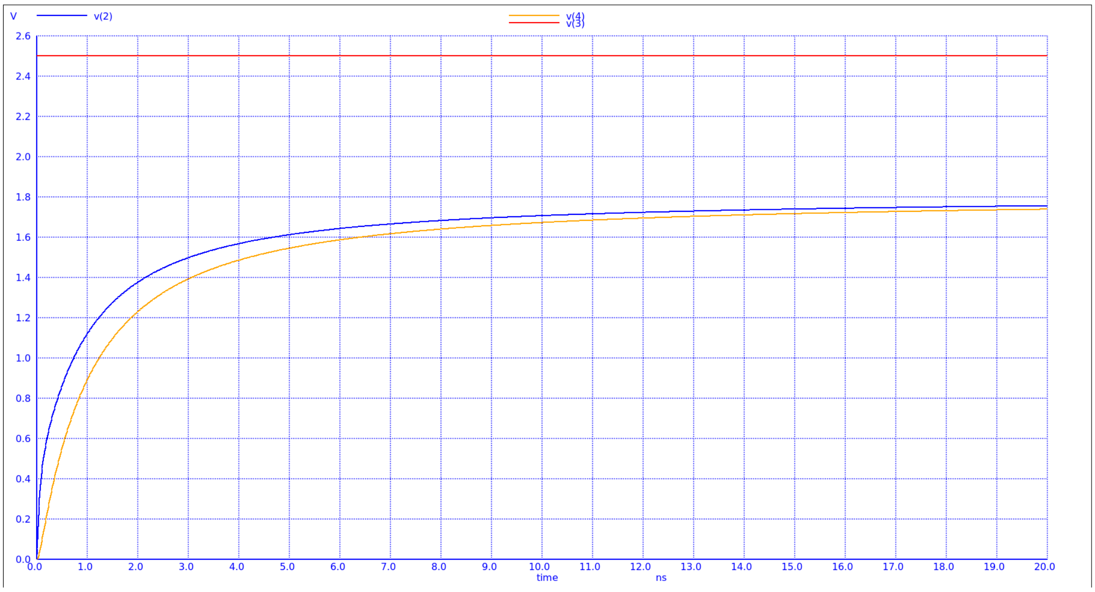


Figure 13: nMOS transistors connected in series with a capacitor

The total voltage drop at the final output is 0.8V. This happens because:

1. M_1 Drop: The first transistor limits the voltage to $V_G - V_{T,eff1}$. Due to the Body Effect ($V_{SB} > 0$), V_T increases, resulting in the $1.7V$ limit.
2. M_2 Behavior: The second transistor does not add an extra threshold drop. Since its gate ($2.5V$) is much higher than its drain ($1.7V$), it operates in the linear region and simply passes the voltage from Node 2 to Node 4.

To conclude, in series NMOS configurations, the maximum output voltage is limited by the first stage's threshold voltage, which is significantly worsened by the Body Effect.

The simulation was repeated for a series configuration of two PMOS transistors. In this case, the devices are evaluated on their ability to pass a logic '0' (GND) to the output.

- Observations: The output voltage at Node 4 does not reach 0V, but stabilizes at a higher potential (approximately $1.1V - 1.0V$ above ground) (14).
- Interpretation: Similar to the NMOS case, the PMOS is a "poor" passer of logic '0'. The first transistor M_1 imposes a voltage drop equal to $|V_{Tp,eff}|$.
- Body Effect:Since the source of the PMOS is not at V_{DD} (where the Bulk is connected), the source-to-bulk potential difference increases the threshold voltage magnitude, resulting in a significant residual voltage at the output.

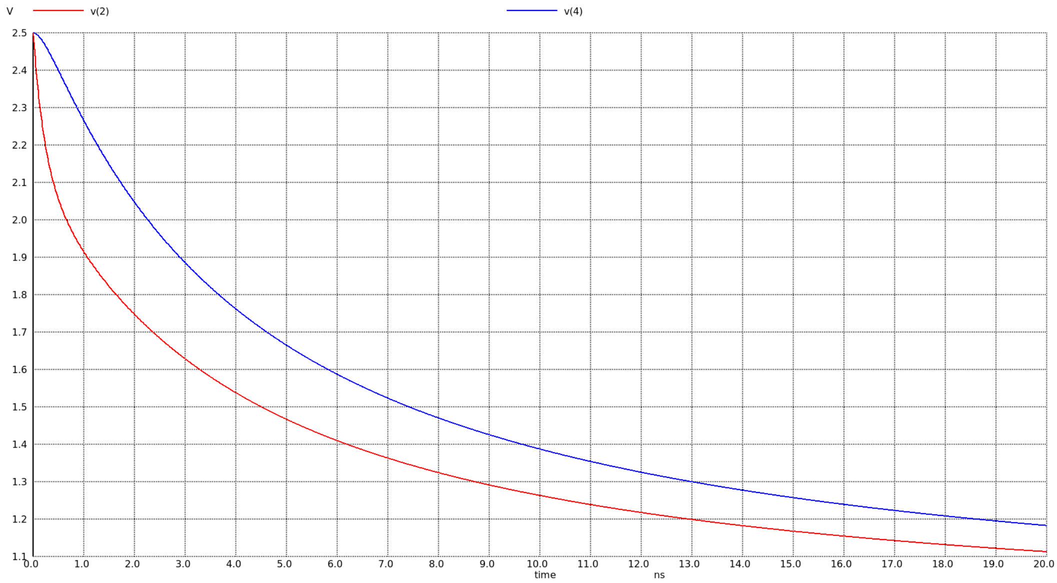


Figure 14: nMOS transistors connected in series with a capacitor

Exercise 3

(a)

Using ngspice, we performed a DC sweep analysis to extract the Voltage Transfer Characteristic (VTC) of a CMOS inverter. The circuit consists of an NMOS and a PMOS transistor, both featuring a channel length of $L = 2\,\mu m$ and a width of $W = 3\,\mu m$. The simulation was conducted with the supply voltage (V_{DD}) set at $2.5V$, using the model parameters corresponding to the TSMC technology node $0.25\,\mu m$. . In addition to the primary VTC plot, we generated the graphical representation of the output voltage derivative with respect to the input voltage (dV_{out}/dV_{in}), which represents the small-signal gain of the inverter (15).

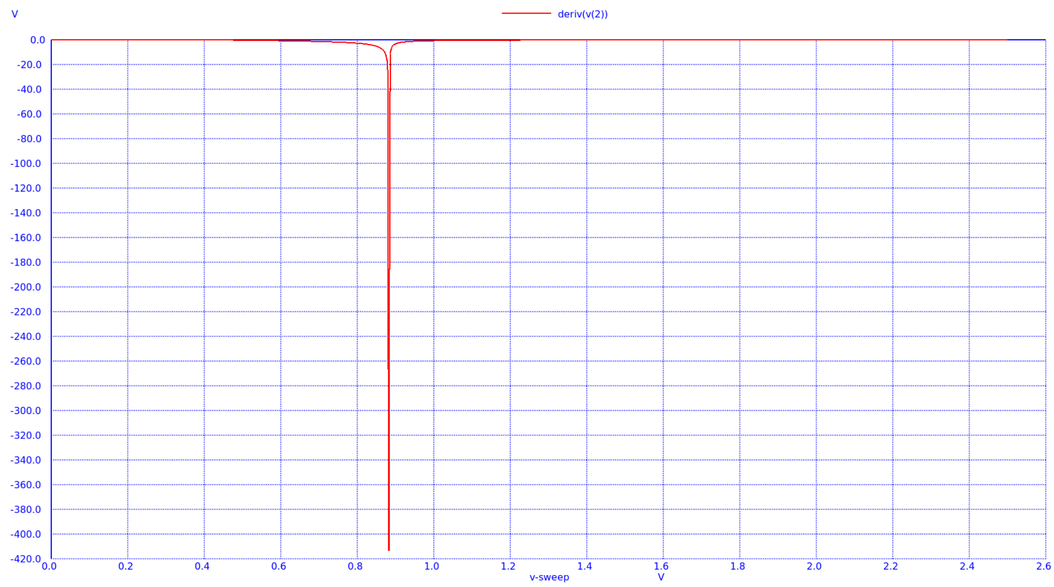


Figure 15: Output voltage derivative graphical representation

The objective of this analysis is to identify the critical input voltage levels, specifically V_{IL}

(Input Low) and V_{IH} (Input High). These points are defined as the coordinates on the VTC where the slope (gain) is exactly equal to -1:

$$\frac{dV_{out}}{dV_{in}} = -1 \tag{3}$$

Determining these points is essential for calculating the Noise Margins (NM_L and NM_H) of the inverter, which characterize its robustness against electrical noise and signal degradation.

The output voltage levels corresponding to these critical points are defined as $V_{OH} = f(V_{IL})$ and $V_{OL} = f(V_{IH})$. Specifically, V_{OH} represents the minimum output voltage that is still logically interpreted as a "1", while V_{OL} denotes the maximum output voltage recognized as a logic "0". These four parameters ($V_{IL}, V_{IH}, V_{OL}, V_{OH}$) constitute the boundaries for the inverter's noise margins. Finally, the Switching Threshold (V_M) is identified as the unique point on the VTC where the input voltage equals the output voltage ($V_{in} = V_{out}$). At this equilibrium point, both transistors typically operate in the saturation region, and the inverter is at its most sensitive state to input variations.The voltage transfer characteristic of the inverter is illustrated in figure 16.

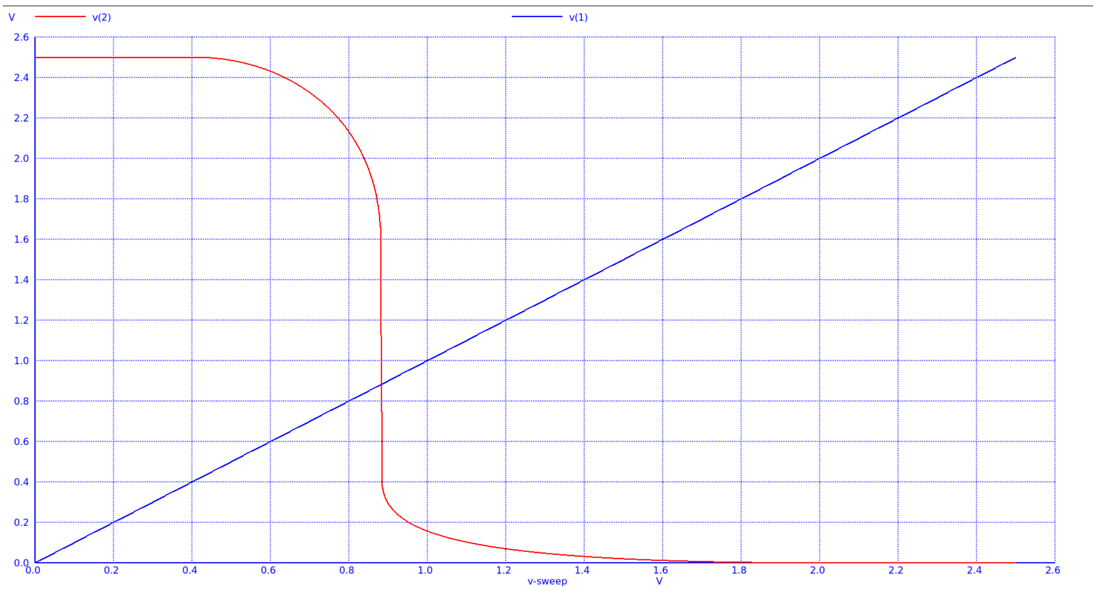


Figure 16: Voltage transfer characteristic of the CMOS inverter

The measured values derived from the previous plots and the subsequent calculation of the Noise Margins are presented below:

Table 8: Critical Voltage Points for the CMOS Inverter

Parameter	**Value (V)**	**Parameter**	**Value (V)**
V_{IL}	0.645	V_{OH}	2.390
V_{IH}	0.964	V_{OL}	0.188
V_M	0.882	V_{DD}	2.500

The Noise Margins for the low (NM_L) and high (NM_H) logic levels are calculated as follows:

$$NM_L = V_{IL} - V_{OL} = 0.645\,V - 0.188\,V = 0.457\,V \tag{4}$$

$$NM_H = V_{OH} - V_{IH} = 2.390\,V - 0.964\,V = 1.426\,V \tag{5}$$

The results indicate a significant asymmetry between the two noise margins ($NM_H > NM_L$). This suggests that the inverter is considerably more robust against noise when the output is at a logic high level compared to a logic low level, which is a direct consequence of the specific transistor sizing and the $0.25\,\mu m$ technology parameters used.

In order to achieve symmetry in the noise margins and equalize the pull-up and pull-down capabilities of the inverter, the switching threshold (V_M) must be shifted to exactly half of the supply voltage ($V_{DD}/2 = 1.25V$). This is accomplished by adjusting the aspect ratios of the transistors (W/L) to compensate for the inherent difference in carrier mobility ($\mu_n > \mu_p$) between the NMOS and PMOS devices.

Based on our iterative simulations, it was observed that for a fixed NMOS width of $W_n = 2\,\mu m$, the switching threshold reaches the ideal value of $V_M = 1.25V$ when the PMOS width increases to $W_p = 12\,\mu m$ (17).

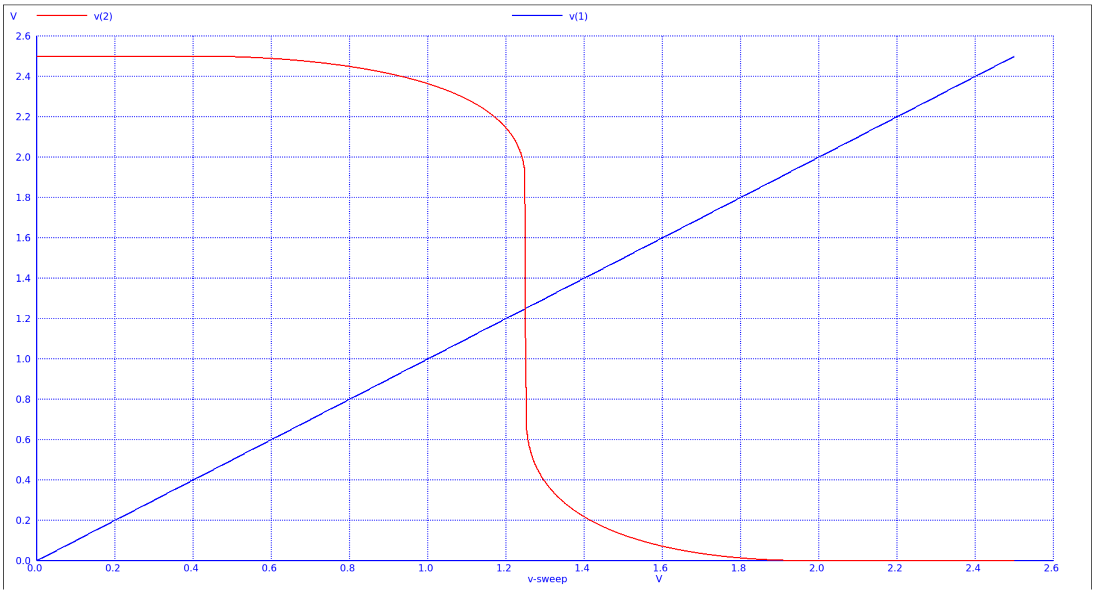


Figure 17: Voltage transfer characteristic of symmetrical CMOS inverter

The results demonstrate that to balance the inverter, the PMOS transistor must be designed approximately six times larger than the NMOS transistor ($W_p \approx 6 \cdot W_n$) in order to lead to optimal and symmetric noise margins, thus enhancing the overall robustness of the digital logic gate.

The new measured data for the symmetric inverter is summarized in Table 9:

Table 9: Critical Voltage Points for the Symmetric Inverter ($W_p/W_n \approx 6$)

Parameter	**Value (V)**	**Parameter**	**Value (V)**
V_{IL}	1.104	V_{OH}	2.280
V_{IH}	1.426	V_{OL}	0.195
V_M	1.249	V_{DD}	2.500

The optimized Noise Margins are calculated as:

$$NM_L = V_{IL} - V_{OL} = 1.1V - 0.195V = 0.905\,V \tag{6}$$

$$NM_H = V_{OH} - V_{IH} = 2.28V - 1.42V = 0.860\,V \tag{7}$$

The simulation results confirm that the noise margins are nearly symmetric. Theoretically, a CMOS inverter achieves a switching threshold of $V_{DD}/2$ when the transconductance ratio is $k_n/k_p = 1$, assuming $V_{tn} = |V_{tp}|$. However, if $k_n/k_p < 1$, the switching threshold shifts above $V_{DD}/2$, while if $k_n/k_p > 1$, it shifts below.

In our case, since $V_{tn} \neq |V_{tp}|$, the ideal symmetry at $V_M = 1.25V$ was not achieved when $k_n/k_p = 1$,in contrast , the value of the ratio is estimated as follows:

$$\frac{k_n}{k_p} = \frac{k'_n(\frac{W_n}{L_n})}{k'_p(\frac{W_p}{L_p})} = \frac{2.501 \cdot 10^{-4} \cdot \frac{2 \cdot 10^{-6}}{2 \cdot 10^{-6}}}{5.194 \cdot 10^{-5} \cdot \frac{12 \cdot 10^{-6}}{2 \cdot 10^{-6}}} = 0.8$$

By further analyzing the transconductance ratio, we have:

$$\frac{k_n}{k_p} = \frac{k'_n(\frac{W_n}{L_n})}{k'_p(\frac{W_p}{L_p})} = \frac{\mu_n C_{ox}(\frac{W_n}{L_n})}{\mu_p C_{ox}(\frac{W_p}{L_p})} = \frac{\mu_n W_n}{\mu_p W_p} \tag{8}$$

Since we now know that $k_n/k_p = 0.8$ and substituting the dimensions $W_n = 2\,\mu m$ and $W_p = 12\,\mu m$ (with $L_n = L_p$), the equation becomes:

$$0.8 = \frac{\mu_n \cdot 2}{\mu_p \cdot 12} \implies \mu_n = \frac{0.8 \cdot 12}{2}\mu_p \implies \mu_n = 4.8\mu_p \tag{9}$$

Bringing everything into consideration the , the current analysis indicates that the electron mobility (μ_n) in this specific $0.25\,\mu m$ TSMC process needs to be approximately 4.8 to 5 times higher than the hole mobility (μ_p) in order to secure the smooth functionality of the circuit. This physical disparity explains why the PMOS transistor must be sized significantly larger than the NMOS to achieve electrical symmetry within the inverter.

(b)

Using **ngspice**, we extracted the Voltage Transfer Characteristics (VTC) of the CMOS inverter across various supply voltage levels ($V_{DD} = 1.8V, 1.2V, 0.7V$). This analysis demonstrates how the inverter's transition region, switching threshold, and overall gain scale with reduced operating voltages. The circuit behavior for each of these values is illustrated in the figures 18,19,20.

It is observed that as the supply voltage decreases, the slope of the Voltage Transfer Characteristic (VTC) in the transition region becomes significantly steeper. This indicates that the inverter gain is inversely proportional to the supply voltage V_{DD}.

Based on our calculations, we quantify the width of the transition region as a percentage of the total supply voltage. The results are summarized below:

- For $V_{DD} = 2.5V$, the transition region width is 12.75% of V_{DD}.
- For $V_{DD} = 1.8V$, it decreases to 9.8%.
- For $V_{DD} = 1.2V$, it further drops to 4.3%.
- For $V_{DD} = 0.7V$, it reaches a minimum of only 2.28%.

According to our observations we can conclude that the substantial reduction in the transition region width at lower supply voltages leads to an improvement in the relative noise margins. By narrowing the ”uncertainty” zone (where the input is neither a clear logic 0 nor a logic 1), the inverter provides a more abrupt switching behavior. This results in larger valid logic 0 and logic 1 input ranges, effectively enhancing the circuit’s robustness against noise at low-voltage operations.

About the power and energy consumption we could say that ,the total energy consumed over a period T is directly proportional to the supply voltage V_{DD}, as defined by the following integral:

$$E = \int_0^T P(t)dt = V_{DD} \int_0^T I_{DD}(t)dt \quad (10)$$

Consequently, reducing the supply voltage leads to a significant decrease in power consumption. Using ngspice, we measured the average static current (I_{avg}) for different V_{DD} values. The results, along with the calculated average power ($P_{avg} = V_{DD} \cdot I_{avg}$), are presented in Table 10.

Table 10: Static Current and Power Consumption vs. V_{DD}

V_{DD} **(V)**	I_{avg} **(Average Current)**	P_{avg} **(Average Power)**
0.7	263 pA	185 pW
1.2	83 nA	98 nW
1.8	1.72 μA	3.1 μW
2.5	6.9 μA	17.5 μW

The data clearly illustrates that lower supply voltages result in drastically lower static current and power dissipation. Specifically, dropping V_{DD} from $2.5V$ to $0.7V$ reduces the power consumption by several orders of magnitude (from microwatts to picowatts). This highlights the fundamental trade-off in digital design: while reducing V_{DD} enhances energy efficiency and power economy, it typically comes at the cost of increased propagation delay.

In conclusion, the scaling of the supply voltage V_{DD} leads to a significant reduction in both static current and overall power consumption, as demonstrated by our measurements. Specifically, lowering V_{DD} achieves energy savings of several orders of magnitude.

Most importantly, our Voltage Transfer Characteristics (VTC) confirm that the CMOS inverter maintains its fundamental switching property even at lower supply levels. While the reduction in voltage increases propagation delay, the inverter continues to function as a robust digital gate, successfully preserving the integrity of logic 0 and logic 1 states.

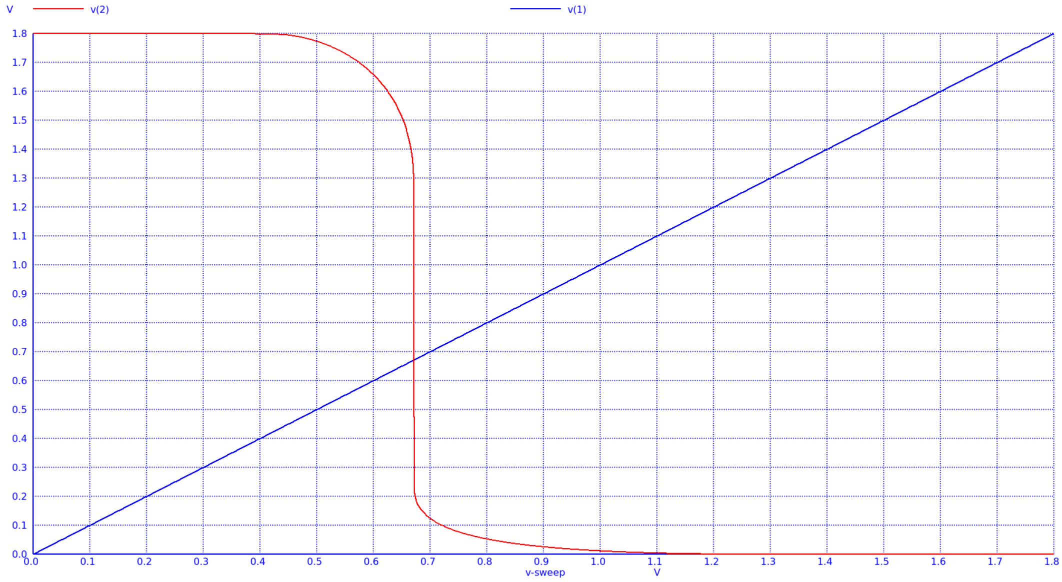


Figure 18: Voltage transfer characteristic of the CMOS inverter for $V_{dd} = 1.8V$

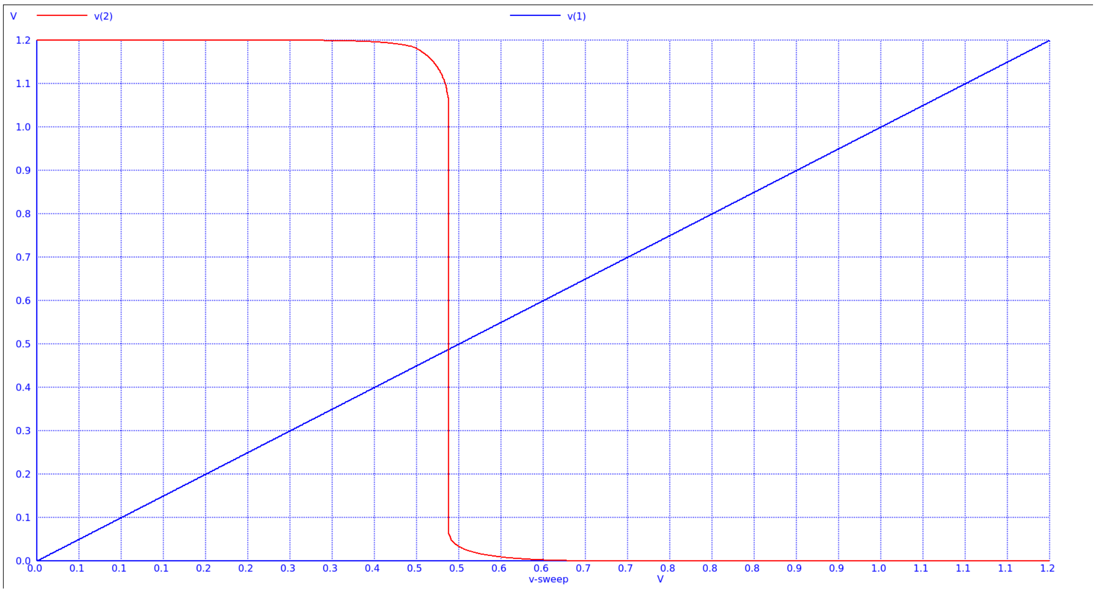


Figure 19: Voltage transfer characteristic of the CMOS inverter for $V_{dd} = 1.2V$

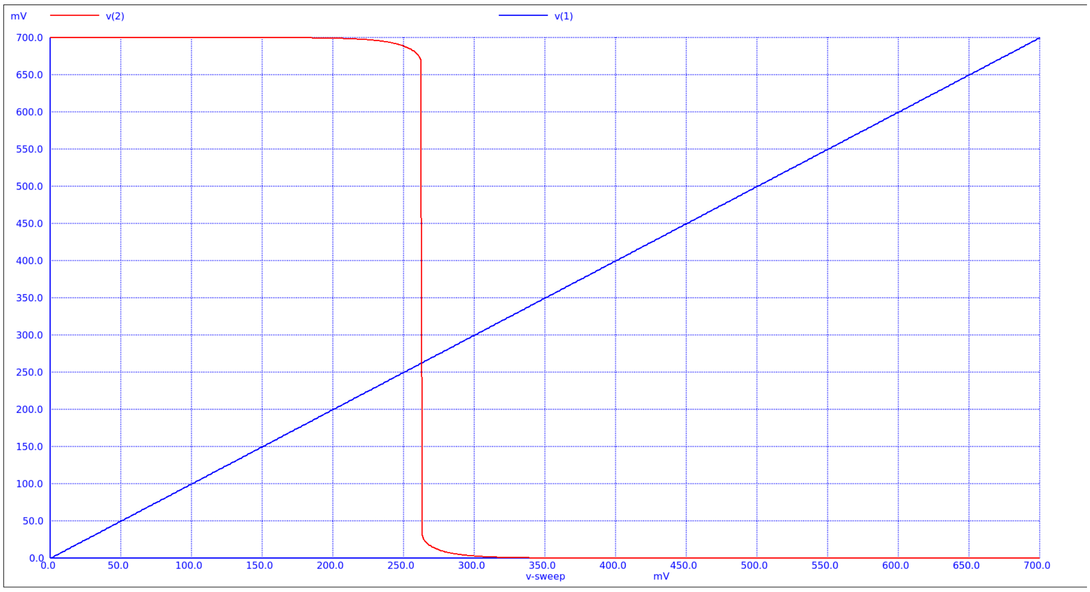


Figure 20: Voltage transfer characteristic of the CMOS inverter for $V_{dd} = 0.7V$